Rejit B Antony

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Education

Vellore Institute of Technology — Chennai, India
Bachelor of Technology – Computer Science Engineering — *July 2023 – Present*
- CGPA: **9.12**

Navdeep Public School — Kollam, Kerala
High School – CBSE — *May 2021 – May 2023*
- Graduated with **94.2%** overall score in Grade 12.

Projects

PICSHARE: AI-Powered Cloud Photo Gallery | *Open-Source Contribution* — GitHub
- Cut **lightbox image load** from **∼2.4 s to <24 ms** (median) by serving pre-generated 2K WebP previews from local disk instead of streaming originals from Google Drive, with files **∼99% smaller** (**30 MB → ∼270 KB**).
- Built a **Drive-backed image store** (Drive as source of truth; 2K WebP previews, thumbnails, and face embeddings on the server), cutting server disk from **∼300 GB** of originals to **∼3.3 GB** per 10k photos.
- Built a TypeScript **Canvas compressor** that shrinks multi-MB selfies to **∼300 KB** before upload, accelerating uploads and avoiding proxy payload errors without hurting **InsightFace** matching.

Reinforcement Learning – BipedalWalker (TD3 & TD3-FORK) | *PyTorch, Gymnasium, CUDA* — GitHub
- Implemented TD3 from scratch in PyTorch (replay buffer, twin critics, actor, Polyak target updates) with no RL library dependencies; trained on BipedalWalker-v3 to avg reward **+280/300** in **1M steps** on personal GPU.
- Built a live **6-panel** Matplotlib training dashboard (reward curves, Q-value drift, action distributions, TD error) and a threaded `pygame` render window with HUD overlay displaying training in real time without blocking the training loop.
- Extended the agent with TD3-FORK (forward dynamics model predicting next state from current state-action) to address long-horizon credit assignment failure on BipedalWalkerHardcore-v3; reward improved from **-40 to +100+** within **900k steps**.

Placement Tracker Platform (Full-Stack Features & Data Pipelines) | *TypeScript, Svelte, SQLite3* — GitHub
- Architected data import pipelines processing **15,000+** records via scripts, ensuring robust **data integrity**.
- Executed a structural database migration from legacy student IDs to a unified **NeoID** architecture, implementing efficient SQL joins and pagination to slash query latency from **2000ms to 50ms**.

Image Denoising Autoencoder | *TensorFlow, Keras, Python* — GitHub
- Built a hybrid Inception-ResNet denoising autoencoder in TensorFlow/Keras trained on DIV2K, using Inception blocks **(1×1, 3×3, 5×5)** and stacked ResNet residual blocks across a **3-stage** encoder.
- Implemented a U-Net style decoder with skip connections and mixed-precision training (`float16`) to preserve spatial detail while halving activation VRAM consumption.
- Engineered a memory-efficient pipeline using a custom Keras `Sequence` generator for on-the-fly noise augmentation across **5000 patches** from **800 source images**.

Skills

Languages: Python, Java, JavaScript, LaTeX
Frameworks & AI: Gemini API, PyTorch, TensorFlow, Keras, Scikit-learn, React, Astro, Node.js, Express.js, Pandas, NumPy
Databases: MongoDB, SQL, SQLite3
Tools: Claude Code, OpenCode, Gemini CLI / AI Agents, Git, VS Code, JupyterLab, Adobe Photoshop, Unity, Godot
Operating Systems: Linux (Arch, WSL, CLI), Windows

Extracurricular Activities

The Linux Club — VIT Chennai
Media Lead — *June 2024 – April 2026*
- Designed and contributed to event frontends. Managed on ground media team. Maintained 100+GB event footage and photos.

The Robotics Club — VIT Chennai
Media & Design Lead — *April 2025 – April 2026*
- Designed visual branding and led live event photo/video coverage.

CodeVolt '25 Hackathon — VIT Chennai
Volunteer — *March 2025*
- Supported event coordination, ensuring a seamless experience for participants and judges and assisted teams with technical queries and troubleshooting during the hackathon.

Certifications

1st Place – FOSSIT (Free and Open-Source Software Event) – LUGVITC, April 2024
Volunteer Recognition Award – Raptee.HV, CodeVolt Hackathon, March 2025
Full Stack MERN Certification (Delta Course) – ApnaCollege
Agile & Design Thinking – IBM
DevOps Fundamentals – IBM